VESIT Green Campus Sustainability Policy 2024-2040

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VESIT GREEN CAMPUS SUSTAINABILITY POLICY (2024-2040)

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Institution: Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Chembur, Mumbai, Maharashtra, India.

Promulgated by: The Governing Board, Academic Council, and the Campus Sustainability Taskforce.

1. EXECUTIVE MANDATE AND STRATEGIC VISION

Vivekananda Education Society's Institute of Technology (VESIT Green Campus), situated in Chembur, Mumbai, formally commits to achieving institutional net-zero operational greenhouse gas emissions by December 31, 2040. As an apex engineering and higher education institution serving over 3,000 undergraduate and postgraduate students alongside 200 faculty and administrative staff across a built-up area of 32,500 m2, the college recognizes that educational institutions must lead by example in practical environmental decarbonisation. This statutory charter establishes binding decarbonisation trajectories, departmental obligations, technological pathways, and continuous monitoring mechanisms governed by the interactive EcoClock AI platform.

The policy is deliberately anchored in the United Nations Sustainable Development Goals (SDGs), establishing verified operational linkages with SDG 6 (Clean Water and Sanitation), SDG 7 (Affordable and Clean Energy), SDG 9 (Industry, Innovation and Infrastructure), SDG 11 (Sustainable Cities and Communities), SDG 12 (Responsible Consumption and Production), SDG 13 (Climate Action), and SDG 15 (Life on Land).

2. JURISDICTIONAL SCOPE AND PHYSICAL BOUNDARIES

This policy exercises mandatory operational jurisdiction over all physical real estate, utility infrastructure, and operational workflows encompassing:

- Academic Block A (BLD_ACA, 7,500 m2, occupancy 1,100), housing the Department of Computer Engineering (DEP_CMPN) and Department of Electronics & Telecommunication (DEP_EXTC).

- Academic Block B (BLD_ACB, 6,800 m2, occupancy 950), accommodating the Department of Information Technology (DEP_INFT).

- Computer Laboratories (BLD_LAB, 5,200 m2, occupancy 650), housing the Department of Artificial Intelligence & Data Science (DEP_AIDS) and high-density computing clusters.

- Library (BLD_LIB, 3,100 m2, occupancy 350), housing print collections and digital repository archives.

- Auditorium (BLD_AUD, 2,800 m2, seating capacity 800), dedicated to institutional conferences and cultural symposia.

- Canteen (BLD_CAN, 2,100 m2, occupancy 400), serving over 2,200 daily meals and snacks.

- Administration Block (BLD_ADM, 5,000 m2, occupancy 200), housing the Principal's Directorate, Finance, and Campus Facilities (DEP_ADMN).

3. QUANTIFIED STRATEGIC TARGETS AND DECARBONISATION BENCHMARKS

The Governing Board mandates five foundational quantitative targets to guide institutional operational investments between baseline year 2024 and 2040:

3.1 Target Net-Zero (TRG_NET_ZERO): Abate the institutional carbon baseline of 2,450,000 kg CO2e (2,450 metric tonnes CO2e/year) to absolute net-zero by December 31, 2040. The abatement pathway mandates a steady reduction of at least 153,125 kg CO2e annually across Scope 1 direct diesel emissions, Scope 2 indirect grid electricity emissions, and Scope 3 commuter transport and landfill waste emissions.

3.2 Target Renewable Penetration (TRG_RENEWABLE): Scale campus clean electricity generation from the 2024 baseline of 14.8% (312,400 kWh/year) to a minimum of 50.0% renewable electricity penetration by December 31, 2030, through on-site rooftop solar photovoltaic generation and green power wheeling agreements.

3.3 Target Zero Waste to Landfill (TRG_ZERO_WASTE): Escalate solid waste diversion from municipal dump sites from 42.0% to 85.0% by December 31, 2028, deploying mandatory source segregation, aerobic composting, and authorized

e-waste recovery.

3.4 Target Water Neutrality (TRG_WATER_NEUTRAL): Curtail annual municipal piped water demand by 30.0% relative to the 2024 baseline of 48,000,000 litres, reducing gross intake to 33,600,000 litres/year by 2029 through integrated rainwater harvesting and greywater recycling.

3.5 Target Green Commuting (TRG_GREEN_COMMUTE): Increase low-carbon commuter modal share from 52.0% to 70.0% of all passenger-kilometres by 2027 by incentivizing suburban rail transit, municipal electric buses, non-motorized cycling, and dedicated campus feeder transit.

4. INSTITUTIONAL INITIATIVES AND ACTION FRAMEWORKS

To operationalize these binding targets, the institution has chartered seven interdisciplinary initiatives:

- INIT_SOLAR_PH1 (Phase-1 Rooftop Solar Photovoltaic Grid-Tied Project): 220 kWp installed monocrystalline arrays across BLD_ACA (80 kWp), BLD_ACB (70 kWp), BLD_LIB (30 kWp), and BLD_ADM (40 kWp), generating 312,400 kWh/year and abating 256,168 kg CO2e.

- INIT_SOLAR_PH2 (Phase-2 Solar Expansion & Battery Storage Proposal): 150 kWp parking canopy arrays and 50 kWp bifacial PV modules on BLD_AUD paired with a 100 kWh LiFePO4 battery energy storage system (BESS) delivering 284,000 kWh/year and avoiding 232,880 kg CO2e.

- INIT_RWH_SYSTEM (Integrated Rainwater Harvesting & Groundwater Recharge): Intercepting 18,000 m2 of terrace catchments across BLD_ACA, BLD_ACB, and BLD_LIB with 450,000 litres of filtration cistern capacity, capturing 12,400,000 litres annually.

- INIT_BIOGAS_CANTEEN (Smart Food Waste Digester & Organic Aerobic Composting Pit): A 250 kg/day biomethanation digester at BLD_CAN displacing 14 commercial LPG cylinders monthly and abating 48,000 kg CO2e.

- INIT_ELECTRIC_SHUTTLE (Zero-Emission Electric Campus Transit & Station Feeder Van): Dedicated electric shuttle vans operating between Chembur/Kurla railway stations and the campus gate, mitigating 36,000 kg CO2e annually.

- INIT_IOT_ENERGY_MGT (Campus-Wide IoT Smart Sub-Metering & Energy Audit Protocol): 45 Modbus-RS485 digital energy meters and edge gateways across seven campus blocks eliminating 98,000 kWh of idle nocturnal baseloads.

- INIT_BIODIVERSITY_ZONE (Miyawaki Urban Native Afforestation & Campus Biodiversity Arboretum): A 600 m2 high-density micro-forest behind BLD_AUD comprising 1,200 native trees sequestering 14,000 kg CO2e/year.

5. GOVERNANCE, VERIFICATION, AND REGULATORY COMPLIANCE

The Campus Sustainability Taskforce, chaired by the Principal and including the Heads of Department of Computer Engineering (Dr. N. Joshi), Information Technology (Dr. S. Nair), Electronics & Telecommunication (Dr. R. Kulkarni), Artificial Intelligence & Data Science (Dr. P. Sharma), and Facilities Director (Mr. V. Deshmukh), shall convene quarterly to audit utility telemetry, verify carbon accounting against documented emission factors (EF_ELEC_GRID 0.82 kg CO2e/kWh and EF_DIESEL_GEN 2.68 kg CO2e/L), and validate compliance with Central Pollution Control Board (CPCB) norms. Full transparency shall be sustained through automated Neo4j property knowledge graph indexing and public-facing visual telemetry.

INSTITUTIONAL AUDIT & COMPLIANCE VERIFICATION

*This statutory record is indexed in the EcoClock AI Knowledge Graph repository under Canonical ID **DOC_POL_001**. All empirical utility telemetry, carbon accounting estimates, and SDG alignments articulated in this document have been reviewed and verified by the VESIT Campus Sustainability Taskforce, Department of Computer Engineering (DEP_CMPN), Department of Information Technology (DEP_INFT), Department of Electronics & Telecommunication (DEP_EXTC), Department of Artificial Intelligence & Data Science (DEP_AIDS), and Campus Facilities (DEP_ADMN). Published by: Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Hashu Advani Memorial Complex, Collector's Colony, Chembur, Mumbai 400074.*